

COMICCRAFT

AI Comic Story Creator using Gemini Models

PHASE 3: Project Design Phase

Overall Architecture

ComicCraft is designed with three primary components: Frontend, Backend, and AI Integration. These components work together to take user input, generate comic content, display it, and export the final result.

Frontend Design

- The interface is developed using HTML, CSS, and Jinja2 templates.
- `index.html` collects story prompt, character name, setting, tone, and art style.
- `comic_preview.html` displays the generated comic panel-by-panel.
- `export_success.html` confirms successful comic export.

Backend Design

- FastAPI manages server-side operations and application routes.
- The backend receives form data or JSON requests from users.
- It coordinates outline generation, story generation, image generation, layout building, and PDF export.
- `routes.py` connects user requests with the corresponding backend functions.

AI Integration Design

- Gemini Flash generates a structured 5-panel comic outline from the user's prompt.
- Gemini Pro expands the outline into detailed narration and character dialogues.
- Stable Diffusion generates comic-style illustrations using the image prompts.

Process Flow

User Input → FastAPI Route → Gemini Flash Outline → Gemini Pro Story → Stable Diffusion Images → Comic Layout Builder → Comic Preview → PDF Export.

Design Goal

The architecture is modular so that frontend interaction, backend processing, AI services, layout creation, and export responsibilities remain clearly separated.

Project documentation prepared for phase-wise submission.